Project Title: **Heart Disease Analysis Using Tableau**

Platform: **SmartBridge / FutureSkills Prime**

Course: **Data Analytics with Tableau**

University: **D.Y.Patil Agriculture & Technical University,Talsande.**

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Organization: **SmartBridge**

Github Repository: **https://github.com/shradha-github/Heart-Disease-Analysis**

Heart Disease Analysis Project

1. Introduction

Heart disease is one of the major causes of death worldwide. Early detection and analysis of risk factors can help in preventing severe health problems. This project focuses on analyzing patient health data to identify patterns and risk indicators associated with heart disease.

The project uses database management, data visualization, and dashboard development techniques to analyze the dataset and present meaningful insights.

2. Problem Statement

Heart disease is one of the leading causes of mortality worldwide. This project analyzes patient health attributes to identify patterns and risk indicators that contribute to heart disease.

3. Project Objectives

The main objectives of this project are:

- To collect and store structured health data in a database
- To clean and prepare the data for analysis
- To identify key risk factors related to heart disease
- To create visualizations for better understanding of data
- To develop an interactive dashboard for data analysis
- To present insights through data storytelling
- To integrate the dashboard into a web interface

4. Tools and Technologies Used

Tool / Technology	Purpose
SQL	Database management and data extraction
Tableau	Data visualization and dashboard creation
Flask	Web application framework for dashboard integration
GitHub	Version control and project management

5. Dataset Description

The dataset contains patient health information used to analyze the risk of heart disease. The important attributes include:

- **HeartDisease** – Indicates whether the patient has heart disease (Yes/No).

- **BMI** – Body Mass Index of the patient, used to measure body fat based on height and weight.

- **Smoking** – Indicates if the patient is a smoker.

- **AlcoholDrinking** – Shows whether the patient consumes alcohol heavily.

- **Stroke** – Indicates if the patient has previously suffered from a stroke.

- **PhysicalHealth** – Number of days in the past month when the patient experienced poor physical health.

- **MentalHealth** – Number of days in the past month when the patient experienced poor mental health.

- **DiffWalking** – Indicates if the patient has difficulty walking or climbing stairs.
- **Sex** – Gender of the patient (Male/Female).
- **AgeCategory** – Age group of the patient.
- **Race** – Ethnicity or race category of the patient.
- **Diabetic** – Indicates if the patient has diabetes.
- **PhysicalActivity** – Shows whether the patient performs regular physical activity.
- **GenHealth** – Self-reported general health condition (Excellent, Very Good, Good, Fair, Poor).
- **SleepTime** – Average number of hours the patient sleeps per day.
- **Asthma** – Indicates if the patient has asthma.
- **KidneyDisease** – Indicates if the patient suffers from kidney disease.
- **SkinCancer** – Indicates if the patient has a history of skin cancer

6. System Architecture

The project workflow includes the following steps:

1. Data collection from heart disease dataset
2. Data storage and management using SQL database
3. Data cleaning and preprocessing using mySQL
4. Data analysis and visualization using Tableau
5. Dashboard creation for interactive insights
6. Integration of dashboard into a web application using Flask

7. Data Processing Steps

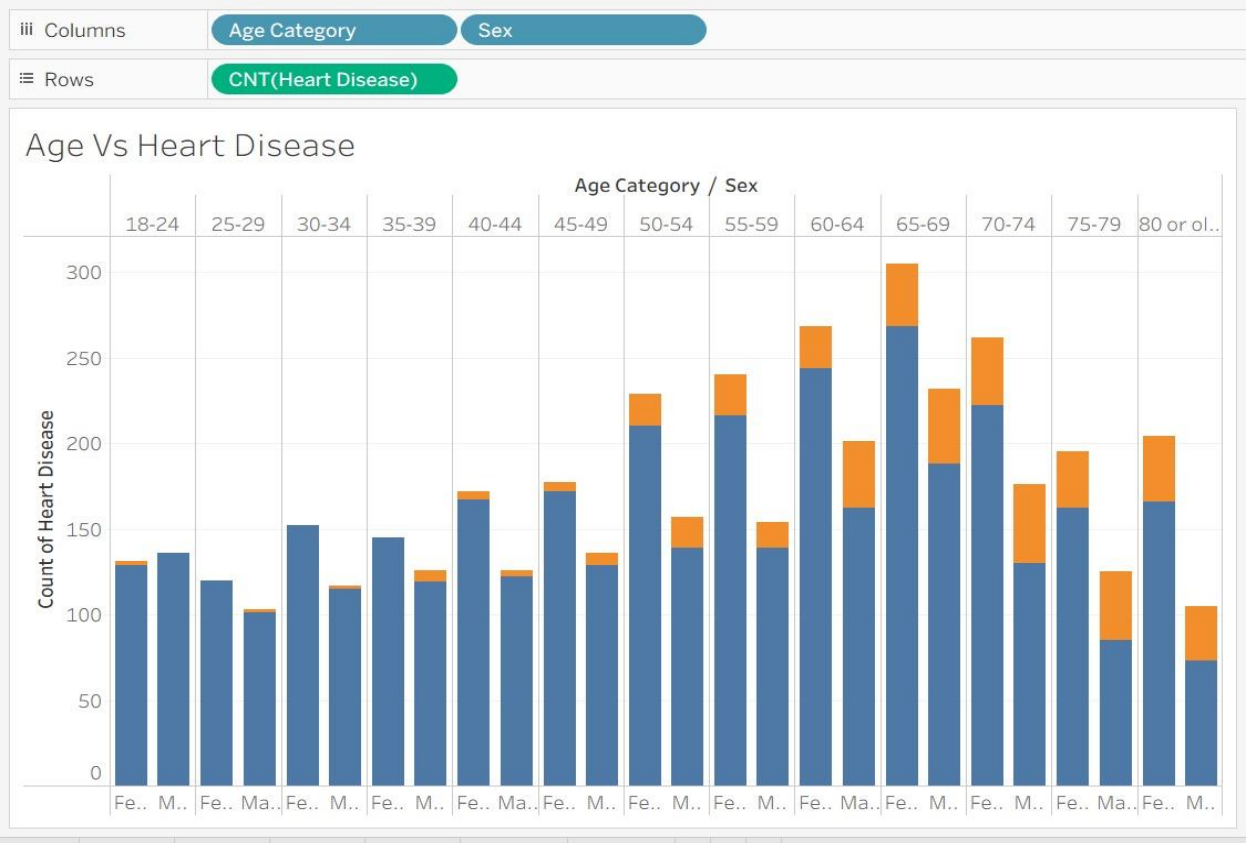
Data Cleaning

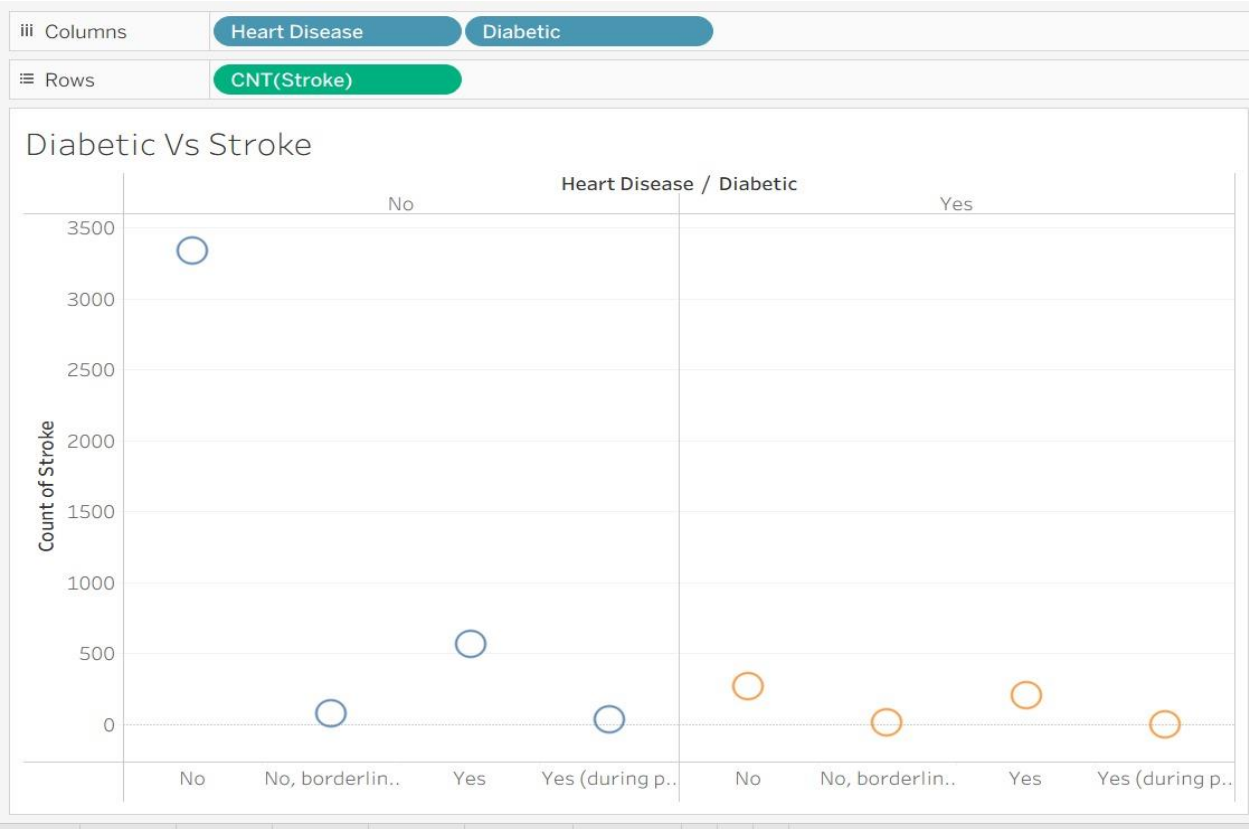
- Removing missing values
- Correcting inconsistent data
- Formatting data for analysis

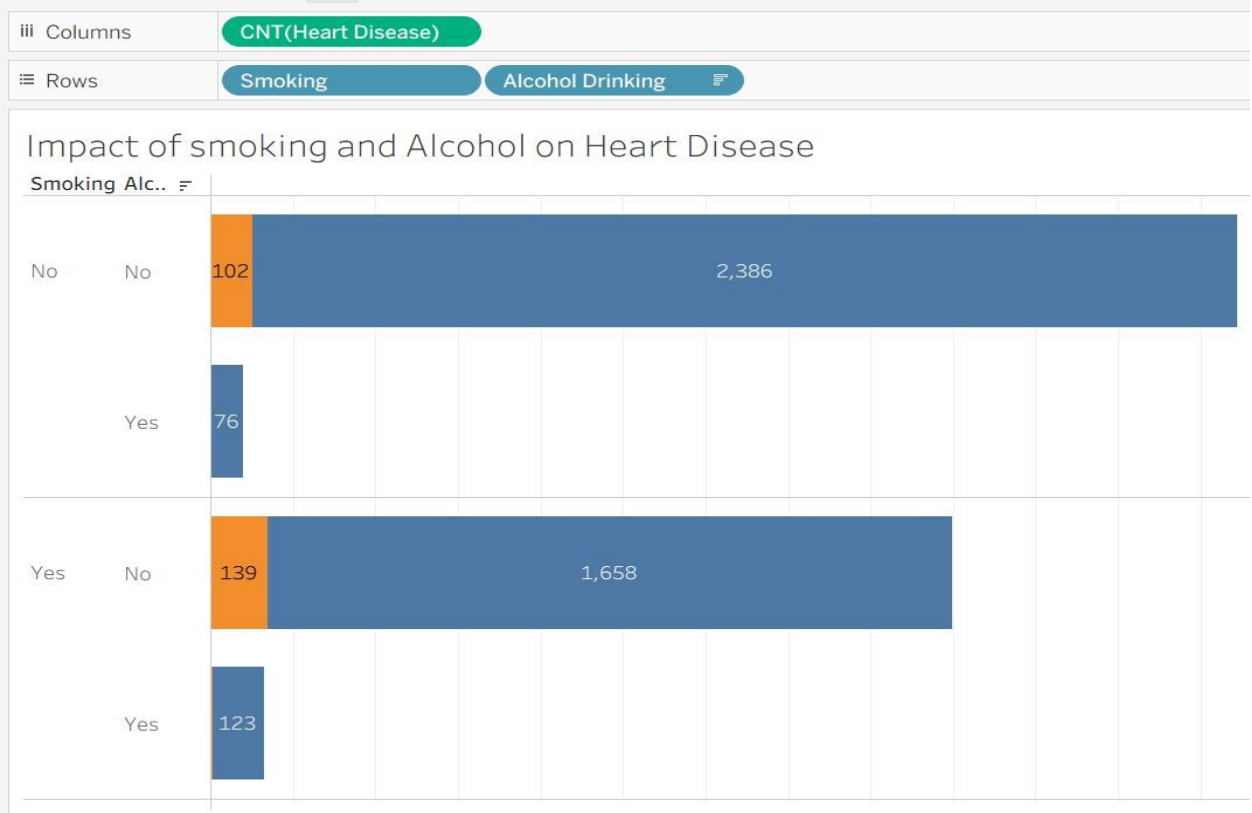
Data Transformation

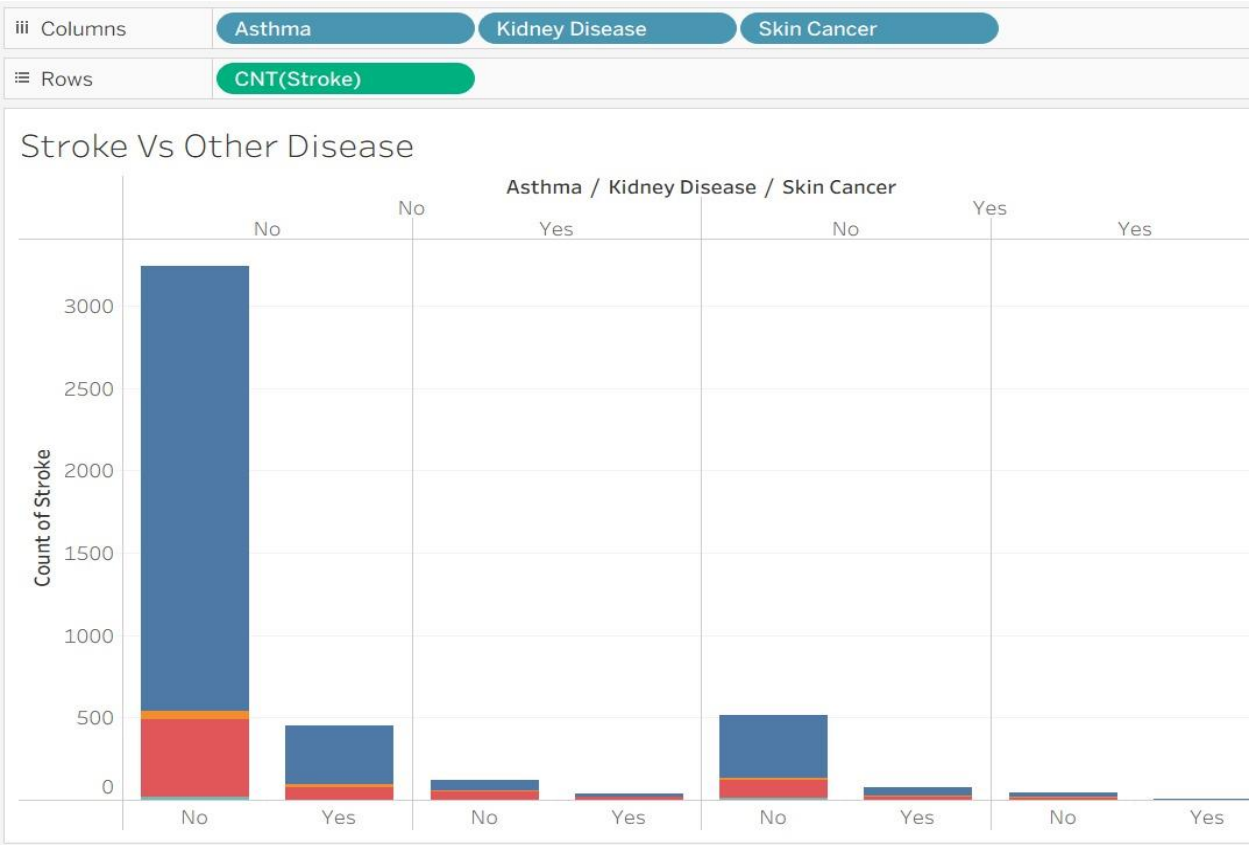
- Converting categorical values
- Normalizing numerical attributes
- Preparing dataset for visualization

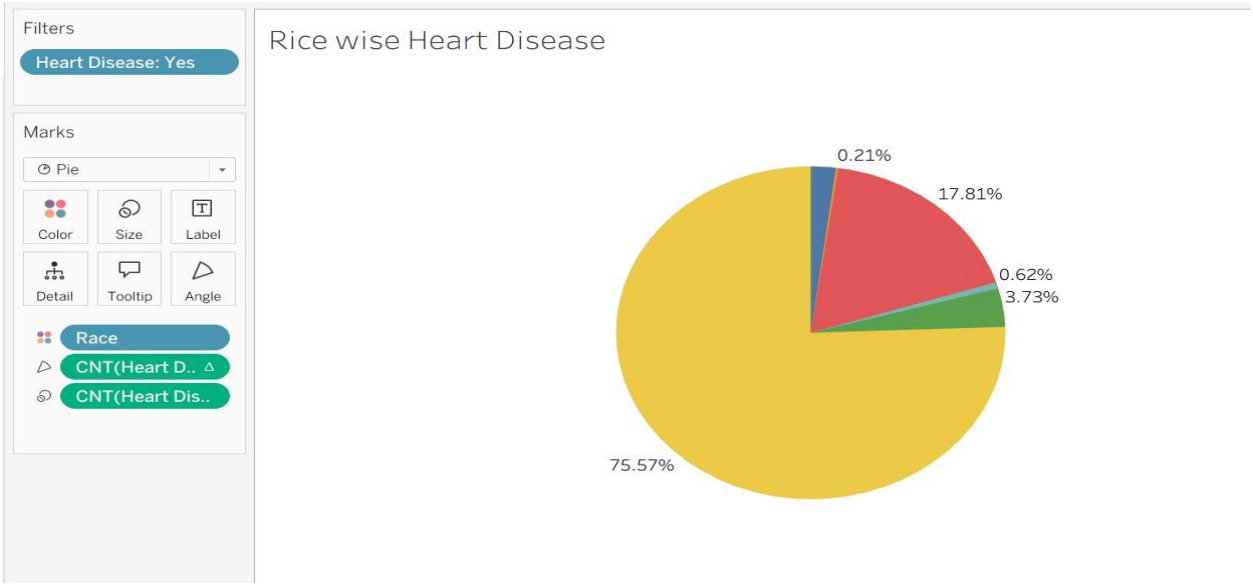
8. Data Visualization

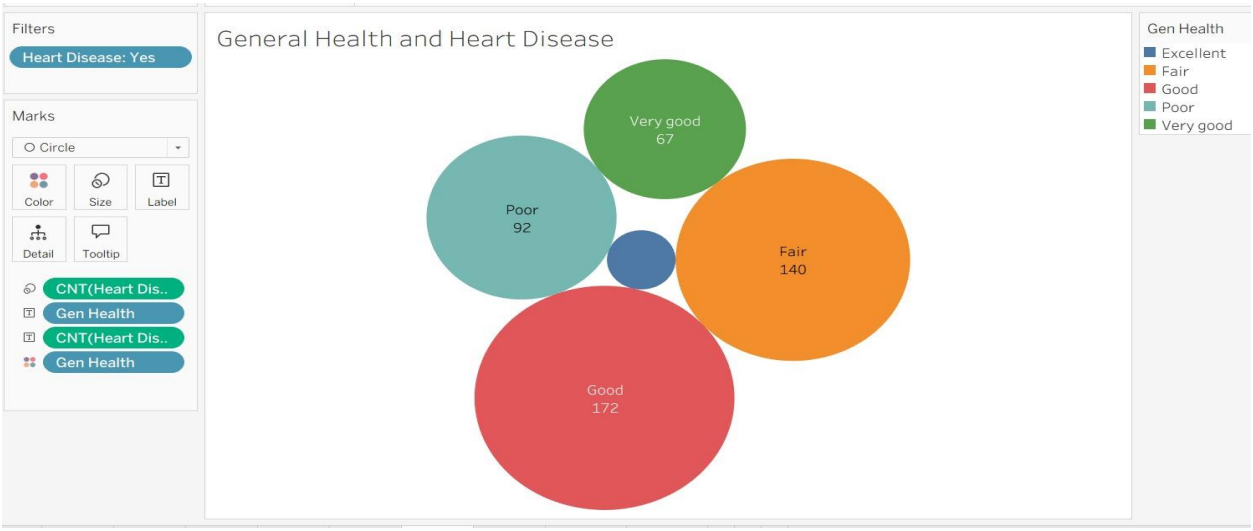

Columns
Age Category
Sex
Rows
CNT(Heart Disease)
Age Vs Heart Disease
Age Category / Sex
18-24
25-29
30-34
35-39
40-44
45-49
50-54
55-59
60-64
65-69
70-74
75-79
80 or ol..
Count of Heart Disease
300
250
200
150
100
50
0
Fe..
M..
Ma..

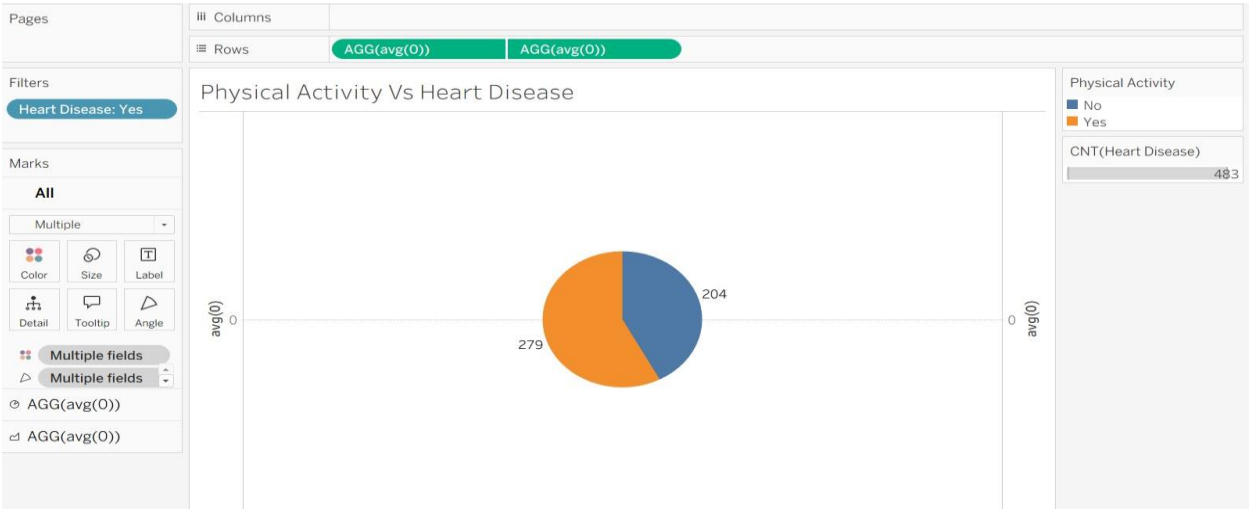

Columns
Heart Disease
Diabetic
Rows
CNT(Stroke)
Diabetic Vs Stroke
Heart Disease / Diabetic
No
Yes
Count of Stroke
3500
3000
2500
2000
1500
1000
500
0
No
No, borderlin..
Yes
Yes (during p..

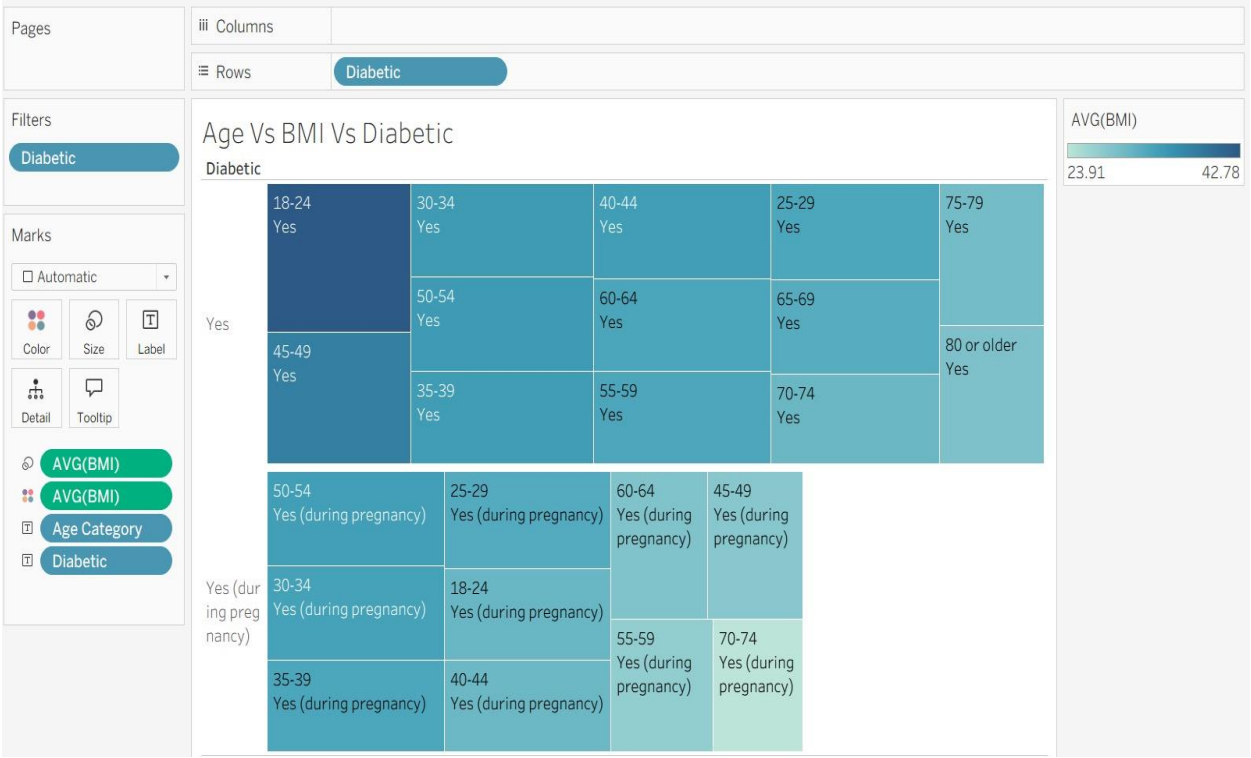

Columns
CNT(Heart Disease)
Rows
Smoking
Alcohol Drinking
Impact of smoking and Alcohol on Heart Disease
Smoking Alc..
No
No
102
2,386
Yes
76
Yes
No
139
1,658
Yes
123

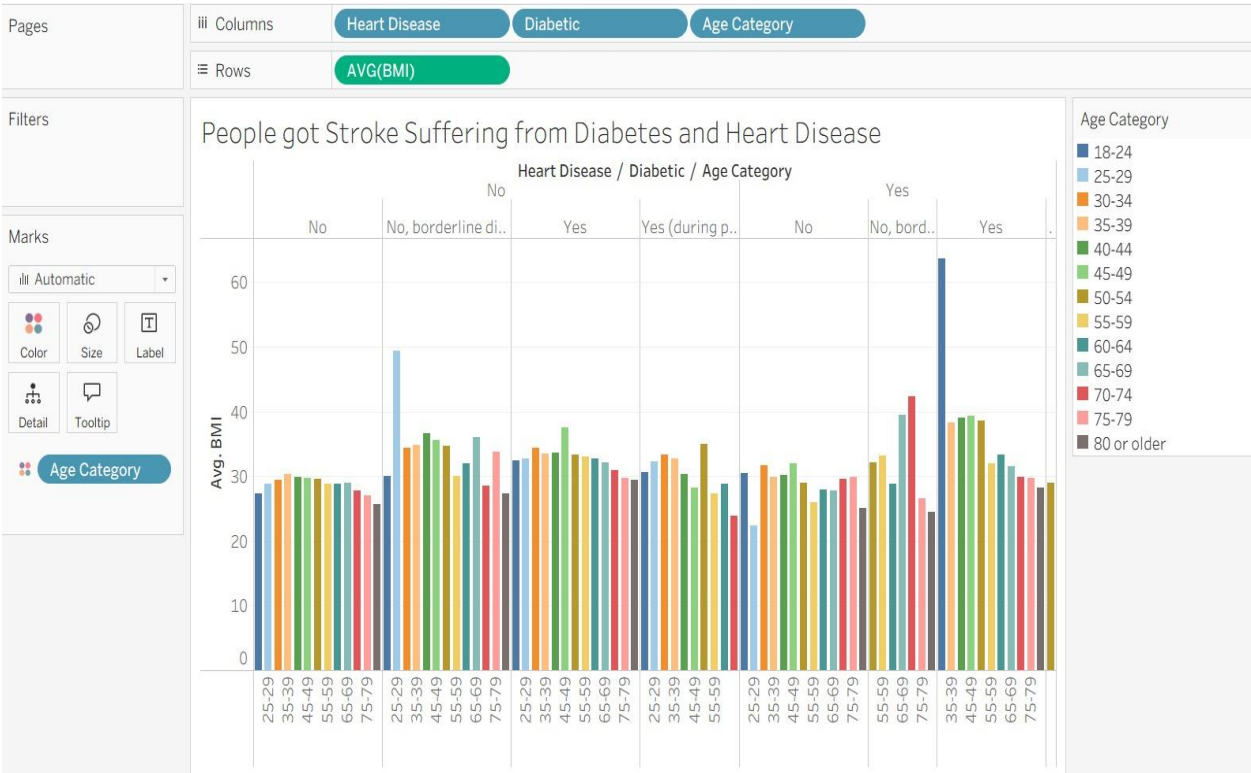

Columns
Asthma
Kidney Disease
Skin Cancer
Rows
CNT(Stroke)
Stroke Vs Other Disease
Asthma / Kidney Disease / Skin Cancer
No
No
Yes
No
Yes
Yes
Count of Stroke
3000
2500
2000
1500
1000
500
0
No
Yes
No
Yes
No
Yes
No
Yes


Filters
Heart Disease: Yes
Marks
Pie
Color
Size
Label
Detail
Tooltip
Angle
Race
CNT(Heart D..
CNT(Heart Dis..
Rice wise Heart Disease
0.21%
17.81%
0.62%
3.73%
75.57%


Filters
Heart Disease: Yes
Marks
Circle
Color
Size
Label
Detail
Tooltip
CNT(Heart Dis..
Gen Health
CNT(Heart Dis..
Gen Health
General Health and Heart Disease
Very good
67
Poor
92
Fair
140
Good
172
Gen Health
Excellent
Fair
Good
Poor
Very good


Pages
Columns
Rows
AGG(avg(0))
AGG(avg(0))
Filters
Heart Disease: Yes
Marks
All
Multiple
Color
Size
Label
Detail
Tooltip
Angle
Multiple fields
Multiple fields
AGG(avg(0))
AGG(avg(0))
Physical Activity Vs Heart Disease
avg(0)
0
204
279
0
avg(0)
Physical Activity
No
Yes
CNT(Heart Disease)
483


Pages
Columns
Rows
Diabetic
Filters
Diabetic
Marks
Automatic
Color
Size
Label
Detail
Tooltip
AVG(BMI)
AVG(BMI)
Age Category
Diabetic
Age Vs BMI Vs Diabetic
Diabetic
Yes
18-24 Yes
30-34 Yes
40-44 Yes
25-29 Yes
75-79 Yes
50-54 Yes
60-64 Yes
65-69 Yes
45-49 Yes
80 or older Yes
35-39 Yes
55-59 Yes
70-74 Yes
Yes (during pregnancy)
50-54 Yes (during pregnancy)
25-29 Yes (during pregnancy)
60-64 Yes (during pregnancy)
45-49 Yes (during pregnancy)
30-34 Yes (during pregnancy)
18-24 Yes (during pregnancy)
55-59 Yes (during pregnancy)
70-74 Yes (during pregnancy)
35-39 Yes (during pregnancy)
40-44 Yes (during pregnancy)
AVG(BMI)
23.91
42.78


Pages
Columns
Heart Disease
Diabetic
Age Category
Rows
AVG(BMI)
Filters
Marks
Automatic
Color
Size
Label
Detail
Tooltip
Age Category
People got Stroke Suffering from Diabetes and Heart Disease
Heart Disease / Diabetic / Age Category
No
Yes
No
No, borderline di..
Yes
Yes (during p..
No
No, bord..
Yes
Avg. BMI
0
10
20
30
40
50
60
25-29
35-39
45-49
55-59
65-69
75-79
Age Category
18-24
25-29
30-34
35-39
40-44
45-49
50-54
55-59
60-64
65-69
70-74
75-79
80 or older

9. Dashboard Development

An interactive dashboard is developed in Tableau to display:

- Patient demographic analysis
- Risk factor analysis
- Comparison charts
- Key performance indicators (KPIs)

The dashboard allows users to interact with filters and explore the data easily.

10. Web Integration

The Tableau dashboard is integrated into a web interface using Python Flask. This allows users to access the dashboard through a browser and interact with the visualizations online.

11. Key Insights

Some important insights from the analysis include:

- Older age groups show higher risk of heart disease
- High cholesterol levels are associated with increased risk
- Certain chest pain types are strong indicators of heart disease

12. Conclusion

This project demonstrates how data analysis and visualization techniques can help in understanding heart disease risk factors. The developed dashboard provides clear insights that can assist healthcare professionals in making better decisions.

13. Future Scope

Future improvements for the project include:

- Adding machine learning models for prediction
- Using larger healthcare datasets
- Improving dashboard interactivity
- Deploying the project on a cloud platform